

4.13.88

AI25BTECH11019 - MENAVATH SAI SANJANA

Question:

A line with positive direction cosines passes through the point $P(2, -1, 2)$ and makes equal angles with the coordinate axes. The line meets the plane $2x + y + z = 9$ at point Q . Find the length of PQ .

Solution:

$$\text{Let } \mathbf{P} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \quad \mathbf{d} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad (0.1)$$

$$\mathbf{r} = \mathbf{P} + t\mathbf{d} = \begin{pmatrix} 2+t \\ -1+t \\ 2+t \end{pmatrix}. \quad (0.2)$$

Plane equation in vector form:

$$\mathbf{n}^T \mathbf{r} = 9, \quad \mathbf{n} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}. \quad (0.3)$$

Intersection point:

$$\mathbf{n}^T (\mathbf{P} + t\mathbf{d}) = 9, \quad (0.4)$$

$$\begin{pmatrix} 2 & 1 & 1 \end{pmatrix} \left(\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right) = 9, \quad (0.5)$$

$$(0.6)$$

$$5 + 4t = 9 \Rightarrow t = 1. \quad (0.7)$$

Coordinates of Q :

$$\mathbf{Q} = \mathbf{P} + t\mathbf{d} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + 1 \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix}. \quad (0.8)$$

Vector PQ and its magnitude:

$$\mathbf{PQ} = \mathbf{Q} - \mathbf{P} = \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad (0.9)$$

$$|PQ| = \sqrt{\mathbf{PQ}^T \mathbf{PQ}} = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}. \quad (0.10)$$

Conclusion: The vector and length of PQ are

$$\mathbf{PQ} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad (0.11)$$

$$|PQ| = \sqrt{3}. \quad (0.12)$$

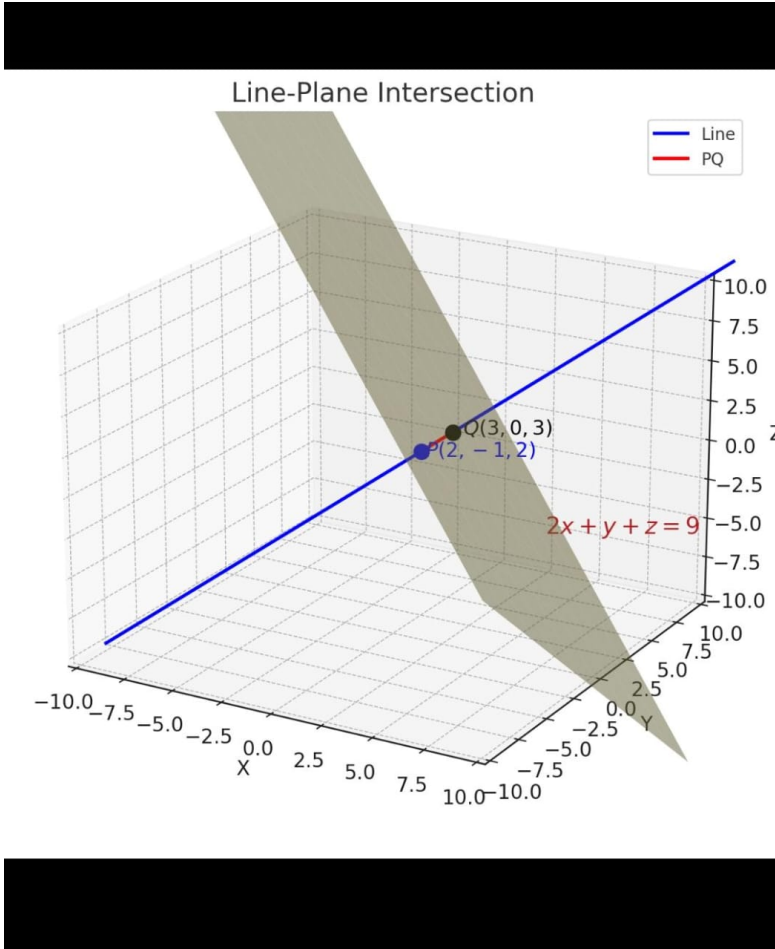


Fig. 0.1